

Result summary

Colour Connect

Vescom BV

Calculation number:	ReTHiNK-80743
Generation on:	22-08-2024
Issue date:	22-08-2024
Valid until:	22-08-2029
Status:	verified

R<THiNK



1 General information

1.1 PRODUCT

Colour Connect

1.2 VALIDITY

Issue date: 22-08-2024

Valid until: 22-08-2029

1.3 OWNER OF THE DECLARATION



Manufacturer: Vescom BV

Address: Sint Jozefstraat 20, 5753 AV Deurne

E-mail: sales@vescom.com

Website: www.vescom.com

Production location: Vescom Textiles GmbH

Address production location: August-Horch-Str. 16, 95213 Münchberg

1.4 VERIFICATION OF THE DECLARATION

The independent verification is in accordance with the ISO 14025:2011. The LCA is in compliance with ISO 14040:2006 and ISO 14044:2006. The EN 15804:2012+A2:2019 serves as the core PCR.

☐ Internal ☒ External



Gert-Jan Vroege, Eco Intelligence

1.5 PRODUCT CATEGORY RULES

EN15804+A2:2019

1.6 FUNCTIONAL UNIT

1m² fabric

1m² fabric to be used as upholstery of furniture.
The fabric weighs 0,290 kg/m² and has a reference service life of 10 years.

Reference unit: square meter (m²)

1.7 CONVERSION FACTORS

Description	Value	Unit
Reference unit	1	m ²
Weight per reference unit	0.290	kg
Conversion factor to 1 kg	3.448276	m ²

1 General information

1.8 SCOPE OF DECLARATION AND SYSTEM BOUNDARIES

This is a Cradle to gate with options LCA. The life cycle stages included are as shown below:

(X = module included, ND = module not declared)

A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
X	X	X	X	X	X	X	X	ND	ND	ND	ND	X	X	X	X	X

The modules of the EN15804 contain the following:

Module A1 = Raw material supply	Module B5 = Refurbishment
Module A2 = Transport	Module B6 = Operational energy use
Module A3 = Manufacturing	Module B7 = Operational water use
Module A4 = Transport	Module C1 = De-construction / Demolition
Module A5 = Construction - Installation process	Module C2 = Transport
Module B1 = Use	Module C3 = Waste Processing
Module B2 = Maintenance	Module C4 = Disposal
Module B3 = Repair	Module D = Benefits and loads beyond the product system boundaries
Module B4 = Replacement	

1.9 COMPARABILITY

In principle, a comparison or assessment of the environmental impacts of different products is only possible if they have been prepared in accordance with EN 15804+A2. For the evaluation of the comparability, the following aspects have to be considered in particular: PCR used, functional or declared unit, geographical reference, the definition of the system boundary, declared modules, data selection (primary or secondary data, background database, data quality), scenarios used for use and disposal phases, and the life cycle inventory (data collection, calculation methods, allocations, validity period). PCRs and general program instructions of different EPD program operators may differ. Comparability needs to be evaluated. For further guidance, see EN 15804+A2 (5.3 Comparability of EPD for construction products) and ISO 14025 (6.7.2 Requirements for comparability).

2 Product

2.1 PRODUCT DESCRIPTION

Vescom produces high-quality interior products for the international project market mainly in the hospitality, healthcare, retail, offices and education segments.

Vescom develops, produces and distributes wall coverings, upholstery and curtain fabrics. The coverings and fabrics are available in a wide variety of materials, structures, textures and colors.

This LCA considers Colour Connect upholstery fabric made from 100% flame-retardant post-consumer recycled polyester.

The reference service life is 10 years.

Technical information

Among others fire resistance, wear resistance and tensile strength tests are done.

- Flame retardancy, according to EN 1021/1-2, BS 5867 crib 5, NF D 60-013, DIN 4102, Ubereinstimmung, UNI 9175 classe 1, Cal TB 117-2013, section 1, IMO 2010 FTP part 8, NFPA 260 class 1.

2.2 DESCRIPTION PRODUCTION PROCESS

The yarn is imported from Italy, already colored. Yarn is delivered to the production site. In the factory the yarn is woven into fabric. The energy needed for these production steps is included in the LCA. After the weaving process, the fabric is washed using washing auxiliaries. After the production process the fabric is inspected manually before packing and shipping. The shipping transport distance is variable and set at 1 km.

2.3 CONSTRUCTION DESCRIPTION

Depending on the final application of the fabric, it must undergo additional processes. These processes should be added separately when the specific processes/values of the project are known, and are left outside the scope of this LCA. Hence, users of the LCA can determine for their respective project what the value must be, and find their sources accordingly. Therefore, apart from the packaging material waste and construction waste, the LCA does not take this module into account, but should be added when the LCA is used in a specific product.

3 Results

3.1 ENVIRONMENTAL IMPACT INDICATORS PER SQUARE METER

CORE ENVIRONMENTAL IMPACT INDICATORS EN15804+A2

Abbr.	Unit	A1	A2	A3	A1- A3	A4	A5	B1	B2	B3	C1	C2	C3	C4	D	Total
AP	mol H+ eqv.	2.76E-3	6.13E-5	7.12E-3	9.94E-3	9.31E-7	6.60E-4	0.00E+0	0.00E+0	0.00E+0	0.00E+0	2.58E-5	2.99E-4	1.88E-6	5.04E-3	1.60E-2
GWP-total	kg CO2 eqv.	5.71E-1	1.06E-2	-1.37E-1	4.44E-1	1.61E-4	1.60E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	4.45E-3	4.97E-1	3.01E-3	1.11E+0	3.65E+0
GWP-b	kg CO2 eqv.	2.76E-3	4.88E-6	-1.39E+0	-1.39E+0	7.41E-8	1.47E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	2.05E-6	-6.07E-5	2.31E-6	-1.26E-2	7.33E-2
GWP-f	kg CO2 eqv.	5.67E-1	1.06E-2	1.24E+0	1.82E+0	1.61E-4	1.23E-1	0.00E+0	0.00E+0	0.00E+0	0.00E+0	4.44E-3	4.97E-1	3.01E-3	1.12E+0	3.57E+0
GWP-luluc	kg CO2 eqv.	1.07E-3	3.87E-6	9.26E-2	9.37E-2	5.88E-8	2.83E-3	0.00E+0	0.00E+0	0.00E+0	0.00E+0	1.63E-6	5.54E-5	1.06E-7	3.41E-4	9.69E-2
EP-m	kg N eqv.	4.61E-4	2.16E-5	1.53E-3	2.01E-3	3.28E-7	1.99E-4	0.00E+0	0.00E+0	0.00E+0	0.00E+0	9.08E-6	8.19E-5	1.14E-6	6.75E-4	2.98E-3
EP-fw	kg P eqv.	2.89E-5	1.07E-7	7.78E-5	1.07E-4	1.62E-9	4.03E-6	0.00E+0	0.00E+0	0.00E+0	0.00E+0	4.48E-8	1.91E-6	3.86E-9	3.77E-5	1.50E-4
EP-T	mol N eqv.	5.29E-3	2.38E-4	2.06E-2	2.62E-2	3.62E-6	2.29E-3	0.00E+0	0.00E+0	0.00E+0	0.00E+0	1.00E-4	9.07E-4	6.90E-6	7.04E-3	3.65E-2
ODP	kg CFC 11 eqv.	2.82E-8	2.33E-9	1.50E-7	1.80E-7	3.54E-11	1.52E-8	0.00E+0	0.00E+0	0.00E+0	0.00E+0	9.80E-10	1.65E-8	6.66E-11	1.01E-7	3.14E-7
POCP	kg NMVOC eqv.	1.41E-3	6.80E-5	4.72E-3	6.19E-3	1.03E-6	6.44E-4	0.00E+0	0.00E+0	0.00E+0	0.00E+0	2.86E-5	2.59E-4	2.64E-6	3.29E-3	1.04E-2
ADP-f	MJ	7.42E+0	1.59E-1	1.88E+1	2.64E+1	2.42E-3	1.40E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	6.70E-2	6.66E-1	5.10E-3	1.82E+1	4.67E+1
ADP-mm	kg Sb- eqv.	5.35E-6	2.68E-7	1.40E-5	1.96E-5	4.07E-9	2.20E-6	0.00E+0	0.00E+0	0.00E+0	0.00E+0	1.13E-7	9.77E-7	2.29E-9	1.56E-5	3.86E-5
WDP		8.80E-2	5.70E-4	4.44E-1	5.32E-1	8.67E-6	2.01E-2	0.00E+0	0.00E+0	0.00E+0	0.00E+0	2.40E-4	2.97E-2	2.18E-4	8.68E-1	1.45E+0

AP=Acidification (AP) | **GWP-total**=Global warming potential (GWP-total) | **GWP-b**=Global warming potential - Biogenic (GWP-b) | **GWP-f**=Global warming potential - Fossil (GWP-f) | **GWP-luluc**=Global warming potential - Land use and land use change (GWP-luluc) | **EP-m**=Eutrophication marine (EP-m) | **EP-fw**=Eutrophication, freshwater (EP-fw) | **EP-T**=Eutrophication, terrestrial (EP-T) | **ODP**=Ozone depletion (ODP) | **POCP**=Photochemical ozone formation - human health (POCP) | **ADP-f**=Resource use, fossils (ADP-f) | **ADP-mm**=Resource use, minerals and metals (ADP-mm) | **WDP**=Water use (WDP)

3 Results

Abbr.	Unit	A1	A2	A3	A1-A3	A4	A5	B1	B2	B3	C1	C2	C3	C4	D	Total
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m3
world
eqv.

AP=Acidification (AP) | **GWP-total**=Global warming potential (GWP-total) | **GWP-b**=Global warming potential - Biogenic (GWP-b) | **GWP-f**=Global warming potential - Fossil (GWP-f) | **GWP-luluc**=Global warming potential - Land use and land use change (GWP-luluc) | **EP-m**=Eutrophication marine (EP-m) | **EP-fw**=Eutrophication, freshwater (EP-fw) | **EP-T**=Eutrophication, terrestrial (EP-T) | **ODP**=Ozone depletion (ODP) | **POCP**=Photochemical ozone formation - human health (POCP) | **ADP-f**=Resource use, fossils (ADP-f) | **ADP-mm**=Resource use, minerals and metals (ADP-mm) | **WDP**=Water use (WDP)

ADDITIONAL ENVIRONMENTAL IMPACT INDICATORS EN15804+A2

Abbr.	Unit	A1	A2	A3	A1-A3	A4	A5	B1	B2	B3	C1	C2	C3	C4	D	Total
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ETP-fw	CTUe	1.01E+1	1.42E-1	4.46E+1	5.48E+1	2.16E-3	3.21E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	5.97E-2	6.20E+0	5.42E-3	3.33E+1	9.75E+1
PM	disease incidence	1.98E-8	9.51E-10	8.29E-8	1.04E-7	1.44E-11	8.71E-9	0.00E+0	0.00E+0	0.00E+0	0.00E+0	4.00E-10	3.23E-9	3.53E-11	4.55E-8	1.62E-7
HTP-c	CTUh	1.56E-10	4.61E-12	1.15E-9	1.31E-9	7.01E-14	1.58E-10	0.00E+0	0.00E+0	0.00E+0	0.00E+0	1.94E-12	1.22E-10	1.42E-13	8.85E-10	2.48E-9
HTP-nc	CTUh	5.74E-9	1.55E-10	2.07E-8	2.66E-8	2.36E-12	2.00E-9	0.00E+0	0.00E+0	0.00E+0	0.00E+0	6.53E-11	2.31E-9	3.53E-12	3.09E-8	6.19E-8
IR	kBq U235 eqv.	2.99E-2	6.68E-4	6.35E-2	9.40E-2	1.01E-5	5.61E-3	0.00E+0	0.00E+0	0.00E+0	0.00E+0	2.81E-4	2.40E-3	1.99E-5	4.02E-2	1.43E-1
SQP	Pt	1.21E+0	1.38E-1	1.58E+2	1.59E+2	2.10E-3	5.07E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	5.81E-2	3.76E-1	1.20E-2	-6.07E+1	1.04E+2

ETP-fw=Ecotoxicity, freshwater (ETP-fw) | **PM**=Particulate Matter (PM) | **HTP-c**=Human toxicity, cancer (HTP-c) | **HTP-nc**=Human toxicity, non-cancer (HTP-nc) | **IR**=Ionising radiation, human health (IR) | **SQP**=Land use (SQP)

CLASSIFICATION OF DISCLAIMERS TO THE DECLARATION OF CORE AND ADDITIONAL ENVIRONMENTAL IMPACT INDICATORS

ILCD classification	Indicator	Disclaimer
ILCD type / level 1	Global warming potential (GWP)	None

3 Results

ILCD classification	Indicator	Disclaimer
ILCD type / level 2	Depletion potential of the stratospheric ozone layer (ODP)	None
	Potential incidence of disease due to PM emissions (PM)	None
	Acidification potential, Accumulated Exceedance (AP)	None
	Eutrophication potential, Fraction of nutrients reaching freshwater end compartment (EP-freshwater)	None
	Eutrophication potential, Fraction of nutrients reaching marine end compartment (EP-marine)	None
	Eutrophication potential, Accumulated Exceedance (EP-terrestrial)	None
	Formation potential of tropospheric ozone (POCP)	None
	Potential Human exposure efficiency relative to U235 (IRP)	1
ILCD type / level 3	Abiotic depletion potential for non-fossil resources (ADP-minerals&metals)	2
	Abiotic depletion potential for fossil resources (ADP-fossil)	2
	Water (user) deprivation potential, deprivation-weighted water consumption (WDP)	2
	Potential Comparative Toxic Unit for ecosystems (ETP-fw)	2
	Potential Comparative Toxic Unit for humans (HTP-c)	2
	Potential Comparative Toxic Unit for humans (HTP-nc)	2
	Potential Soil quality index (SQP)	2
Disclaimer 1 – This impact category deals mainly with the eventual impact of low dose ionizing radiation on human health of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure nor due to radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, from radon and from some construction materials is also not measured by this indicator.		
Disclaimer 2 – The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.		

3 Results

3.2 INDICATORS DESCRIBING RESOURCE USE AND ENVIRONMENTAL INFORMATION BASED ON LIFE CYCLE INVENTORY (LCI)

PARAMETERS DESCRIBING RESOURCE USE

Abbr.	Unit	A1	A2	A3	A1- A3	A4	A5	B1	B2	B3	C1	C2	C3	C4	D	Total
PERE	MJ	8.27E-1	2.00E-3	2.03E+1	2.11E+1	3.03E-5	6.53E-1	0.00E+0	0.00E+0	0.00E+0	0.00E+0	8.39E-4	5.17E-2	9.02E-5	-1.06E+1	1.12E+1
PERM	MJ	0.00E+0	0.00E+0	1.40E+1	1.40E+1	0.00E+0	4.21E-1	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	1.45E+1
PERT	MJ	8.27E-1	2.00E-3	3.43E+1	3.51E+1	3.03E-5	1.07E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	8.39E-4	5.17E-2	9.02E-5	-1.06E+1	2.56E+1
PENRE	MJ	7.92E+0	1.69E-1	1.91E+1	2.72E+1	2.57E-3	1.48E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	7.11E-2	7.08E-1	5.42E-3	1.33E+1	4.27E+1
PENRM	MJ	8.89E+0	0.00E+0	1.76E+0	1.07E+1	0.00E+0	2.99E-1	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	6.15E+0	1.71E+1
PENRT	MJ	1.68E+1	1.69E-1	2.09E+1	3.79E+1	2.57E-3	1.78E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	7.11E-2	7.08E-1	5.42E-3	1.94E+1	5.99E+1
SM	Kg	2.89E-1	0.00E+0	2.31E-2	3.12E-1	0.00E+0	9.36E-3	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	3.21E-1
RSF	MJ	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
NRSF	MJ	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
FW	M3	4.05E-3	1.94E-5	1.42E-2	1.83E-2	2.95E-7	8.00E-4	0.00E+0	0.00E+0	0.00E+0	0.00E+0	8.16E-6	8.63E-4	5.32E-6	2.30E-2	4.30E-2

PERE=renewable primary energy ex. raw materials | **PERM**=renewable primary energy used as raw materials | **PERT**=renewable primary energy total | **PENRE**=non-renewable primary energy ex. raw materials | **PENRM**=non-renewable primary energy used as raw materials | **PENRT**=non-renewable primary energy total | **SM**=use of secondary material | **RSF**=use of renewable secondary fuels | **NRSF**=use of non-renewable secondary fuels | **FW**=use of net fresh water

OTHER ENVIRONMENTAL INFORMATION DESCRIBING WASTE CATEGORIES

Abbr.	Unit	A1	A2	A3	A1- A3	A4	A5	B1	B2	B3	C1	C2	C3	C4	D	Total
HWD	Kg	4.62E-6	4.04E-7	5.52E-5	6.02E-5	6.14E-9	3.60E-6	0.00E+0	0.00E+0	0.00E+0	0.00E+0	1.70E-7	1.17E-6	7.73E-9	6.64E-6	7.18E-5
NHWD	Kg	3.83E-2	1.01E-2	1.58E-1	2.07E-1	1.54E-4	2.95E-2	0.00E+0	0.00E+0	0.00E+0	0.00E+0	4.25E-3	2.36E-2	2.03E-2	6.80E-2	3.52E-1
RWD	Kg	2.11E-5	1.05E-6	6.68E-5	8.90E-5	1.59E-8	6.65E-6	0.00E+0	0.00E+0	0.00E+0	0.00E+0	4.40E-7	2.47E-6	3.02E-8	3.30E-5	1.32E-4

HWD=hazardous waste disposed | **NHWD**=non hazardous waste disposed | **RWD**=radioactive waste disposed

3 Results

ENVIRONMENTAL INFORMATION DESCRIBING OUTPUT FLOWS

Abbr.	Unit	A1	A2	A3	A1- A3	A4	A5	B1	B2	B3	C1	C2	C3	C4	D	Total
CRU	Kg	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0
MFR	Kg	0.00E+0	0.00E+0	5.26E-3	5.26E-3	0.00E+0	6.98E-1	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	9.57E-2	0.00E+0	0.00E+0	7.99E-1
MER	Kg	0.00E+0	0.00E+0	2.56E-4	2.56E-4	0.00E+0	7.67E-6	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	2.63E-4
EET	MJ	0.00E+0	0.00E+0	-2.08E-1	-2.08E-1	0.00E+0	2.09E-4	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	-2.85E+0	-3.06E+0
EEE	MJ	0.00E+0	0.00E+0	-1.21E-1	-1.21E-1	0.00E+0	1.21E-4	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	0.00E+0	-1.66E+0	-1.78E+0

CRU=Components for re-use | **MFR**=Materials for recycling | **MER**=Materials for energy recovery | **EET**=Exported Energy Thermic | **EEE**=Exported Energy Electric

3 Results

3.3 INFORMATION ON BIOGENIC CARBON CONTENT PER SQUARE METER

BIOGENIC CARBON CONTENT

The following Information describes the biogenic carbon content in (the main parts of) the product at the factory gate per square meter:

Biogenic carbon content	Amount	Unit
Biogenic carbon content in the product	0	kg C
Biogenic carbon content in accompanying packaging	0.4009	kg C

UPTAKE OF BIOGENIC CARBON DIOXIDE

The following amount carbon dioxide uptake is taken into account. Related uptake and release of carbon dioxide in downstream processes are not taken into account in this number although they do appear in the presented results.

Uptake Biogenic Carbon dioxide	Amount	Unit
Packaging	1.47	kg CO2 (biogenic)

4 Contact information

Publisher	Operator	Owner of declaration
 Vescom BV Sint Jozefstraat 20 5753 AV Deurne, NL E-mail: sales@vescom.com Website: www.vescom.com	 Vescom BV Sint Jozefstraat 20 5753 AV Deurne, NL E-mail: sales@vescom.com Website: www.vescom.com	 Vescom BV Sint Jozefstraat 20 5753 AV Deurne, NL E-mail: sales@vescom.com Website: www.vescom.com